

9.1.5 Power Metrology Applications

Each channel of the ADS131M02 is identical, giving designers the flexibility to sense voltage or current with any channel. Simultaneous sampling allows the application to calculate instantaneous power for any simultaneous voltage and current measurement. This section provides several diagrams depicting the common energy metrology configurations that can be used with the ADS131M02. A Rogowski coil can alternatively be used to sense current in the following examples wherever a CT is used. The integration to determine the current flowing through the Rogowski coil is done digitally if that modification is made. RC antialiasing filters are not shown in the following diagrams for simplicity, but are recommended for all channels.

Figure 9-4 shows a single phase configuration where the live current is monitored using a CT and the live phase voltage is measured using a voltage divider.

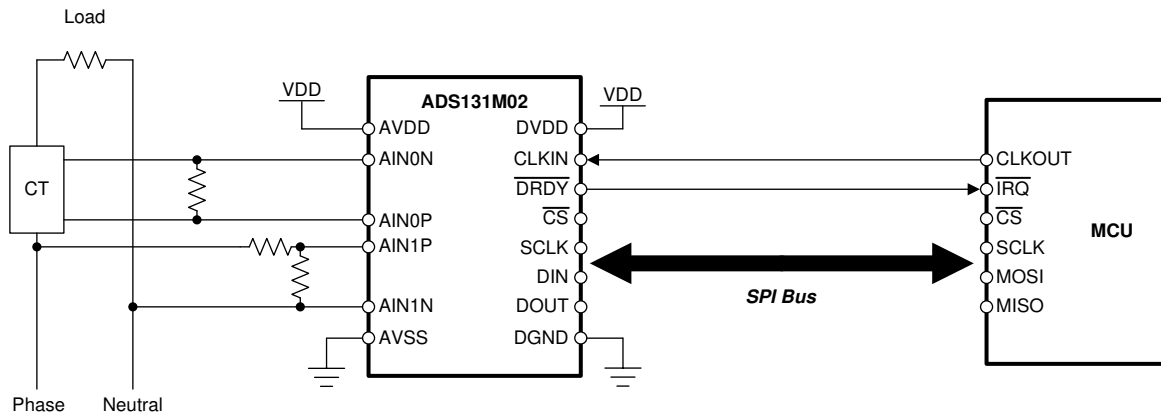


Figure 9-4. Single-Phase CT for Live and Neutral Currents, With Phase Voltage Measured